

Software Engineering Lab UE24CS341A

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

PROBLEM STATEMENT 6: ALUMNI MENTORSHIP & MOCK INTERVIEW PLATFORM

Name: HARSHITHA.B.T	SRN: PES1UG24AM118	Section: AIML - “B”
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1. Introduction & Scenario Review

The Alumni Mentorship & Mock Interview Platform is designed to connect students with alumni mentors working in specialized domains. It supports domain-based mentor matching, mentor availability management, session booking, mock-interview track selection, and structured post-interview feedback and experience sharing.

The main stakeholders identified in Lab 1 are the Student Mentee and Alumni Mentor.

The Student Mentee uses the platform to find suitable mentors, view availability, and schedule mentorship or mock-interview sessions.

The Alumni Mentor provides expertise and availability, conducts mentoring or mock-interview sessions, and provides structured feedback.

Lab 2 converts the functional requirements identified in Lab 1 into Agile Epics and User Stories, prioritizes and estimates the backlog, and uses the Jira Scrum board to simulate two one-week sprints.

2. Elicit & Draft Agile Requirements

The requirements were grouped into major functional themes and decomposed into end-user-focused User Stories.

Requirement	Lab 1 Functional Requirement	Agile Epic	Priority
FR-001	Recommend suitable alumni mentors based on the student's selected domain of interest.	Domain-Based Mentor Matching	High
FR-002	Allow alumni mentors to create, update, and remove available mentoring time slots.	Mentor Availability & Scheduling	High
FR-003	Allow students to book mentorship/mock-interview sessions and select an interview domain or track.	Mock Interview Session Management	High
FR-004	Allow alumni mentors to submit a structured post-interview scorecard containing ratings and comments.	Structured Interview Assessment	High
FR-005	Allow alumni mentors to publish interview experience posts containing preparation methods and relevant information.	Alumni Interview Experience Sharing	Medium

2.1 Agile Epics

The following five Epics were created and derived from the Lab 1 requirements.

Jira ID	Epic	Traceability	Priority
AMPSE-1	Domain-Based Mentor Matching	FR-001	High
AMPSE-2	Mentor Availability & Scheduling	FR-002	High
AMPSE-3	Mock Interview Session Management	FR-003	High
AMPSE-4	Structured Interview Assessment	FR-004	High
AMPSE-5	Alumni Interview Experience Sharing	FR-005	Medium

Jira Epic creation screenshot:

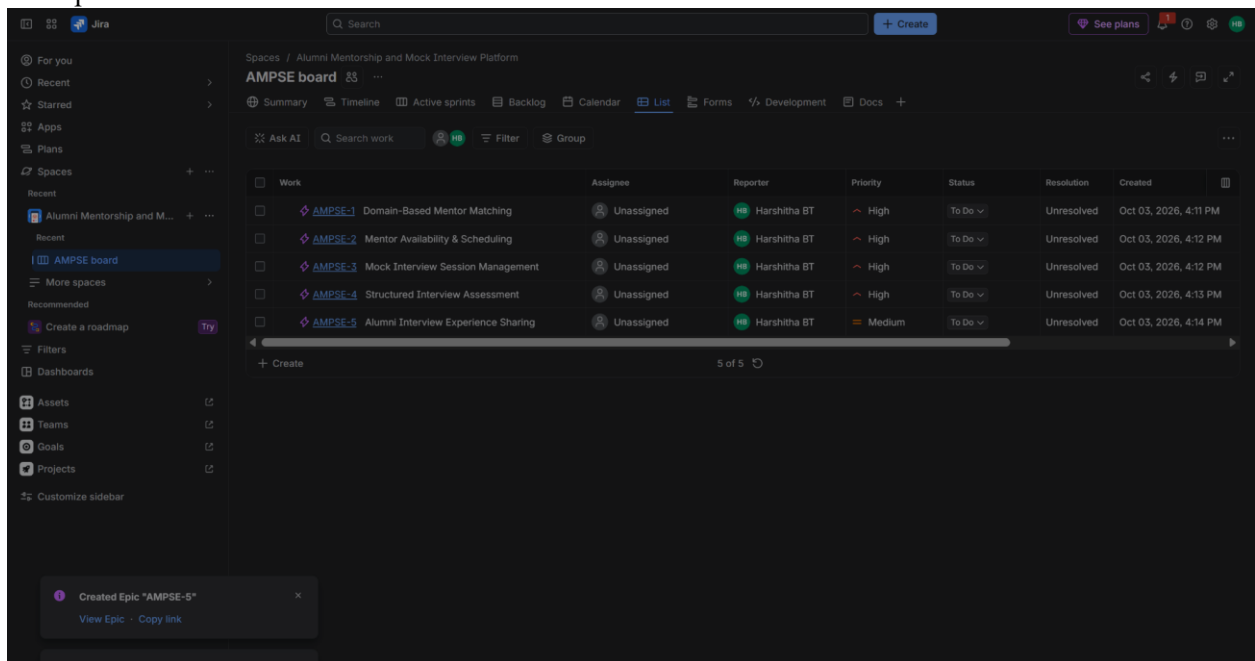


Figure 1 : Jira backlog showing the five created Epics with their priorities.

3. User Stories, Priorities & Story Points

For each Epic, related User Stories were written using the required Agile format:

“As a [user role], I want [goal], so that [benefit]”

Priorities were assigned based on user impact. Story Points were estimated using the Fibonacci-style scale, considering relative effort, complexity, and uncertainty.

ID	User Story	Epic	User Story Statement	Priority	Points
AMPSE-6	Select Mentor Domain	Domain-Based Mentor Matching	As a student, I want to select my domain of interest, so that I can receive mentor recommendations relevant to my needs.	High	3
AMPSE-7	Receive Domain-Based Mentor Recommendations	Domain-Based Mentor Matching	As a student, I want the system to recommend suitable alumni mentors based on my selected domain, so that I can find relevant mentors efficiently.	High	5
AMPSE-8	View Recommended Mentor Details	Domain-Based Mentor Matching	As a student, I want to view the expertise and relevant details of recommended mentors, so that I can make an informed choice.	High	3

AMPSE-9	Create Available Time Slot	Mentor Availability & Scheduling	As an alumni mentor, I want to create available mentoring time slots, so that students can schedule sessions with me.	High	3
AMPSE-10	Update Available Time Slot	Mentor Availability & Scheduling	As an alumni mentor, I want to update my available time slots, so that students see my current availability.	High	3
AMPSE-11	Remove Available Time Slot	Mentor Availability & Scheduling	As an alumni mentor, I want to remove an available time slot, so that unavailable periods cannot be booked by students.	High	3
AMPSE-12	Book Mentorship or Mock Interview Session	Mock Interview Session Management	As a student, I want to book a mentorship or mock-interview session with a selected mentor, so that I can receive guidance or practice an interview.	High	5
AMPSE-13	Select Interview Domain or Track	Mock Interview Session Management	As a student, I want to select an interview domain or track such as DSA, System Design, or HR, so that the session is focused on my chosen area.	High	3
AMPSE-14	Prevent Double Booking	Mock Interview Session Management	As a student, I want the system to prevent booking an already-booked mentor time slot, so that I can schedule a session without conflicts.	High	5
AMPSE-15	Submit Interview Scorecard	Structured Interview Assessment	As an alumni mentor, I want to submit a structured post-interview scorecard with ratings and comments, so that I can provide meaningful feedback to the student.	High	5
AMPSE-16	Validate Scorecard Submission	Structured Interview Assessment	As an alumni mentor, I want the system to validate required ratings and comments before submission, so that incomplete feedback is not recorded.	High	3
AMPSE-17	View Interview Feedback	Structured Interview Assessment	As a student, I want to view the submitted interview scorecard, so that I can understand my performance and areas for improvement.	High	3
AMPSE-18	Publish Interview Experience	Alumni Interview Experience Sharing	As an alumni mentor, I want to publish an interview experience post, so that students can learn from my interview experience.	Medium	5
AMPSE-19	Share Interview Preparation Methods	Alumni Interview Experience Sharing	As an alumni mentor, I want to include preparation methods and relevant information in my experience post, so that students can use the information for interview preparation.	Medium	3
AMPSE-20	Control Contact Information Visibility	Alumni Interview Experience Sharing	As an alumni mentor, I want my contact information in an interview experience post to follow my privacy settings, so that my personal information is not exposed to unauthorized users.	Medium	3

Backlog with Epics, User Stories, Priorities and Story Points:

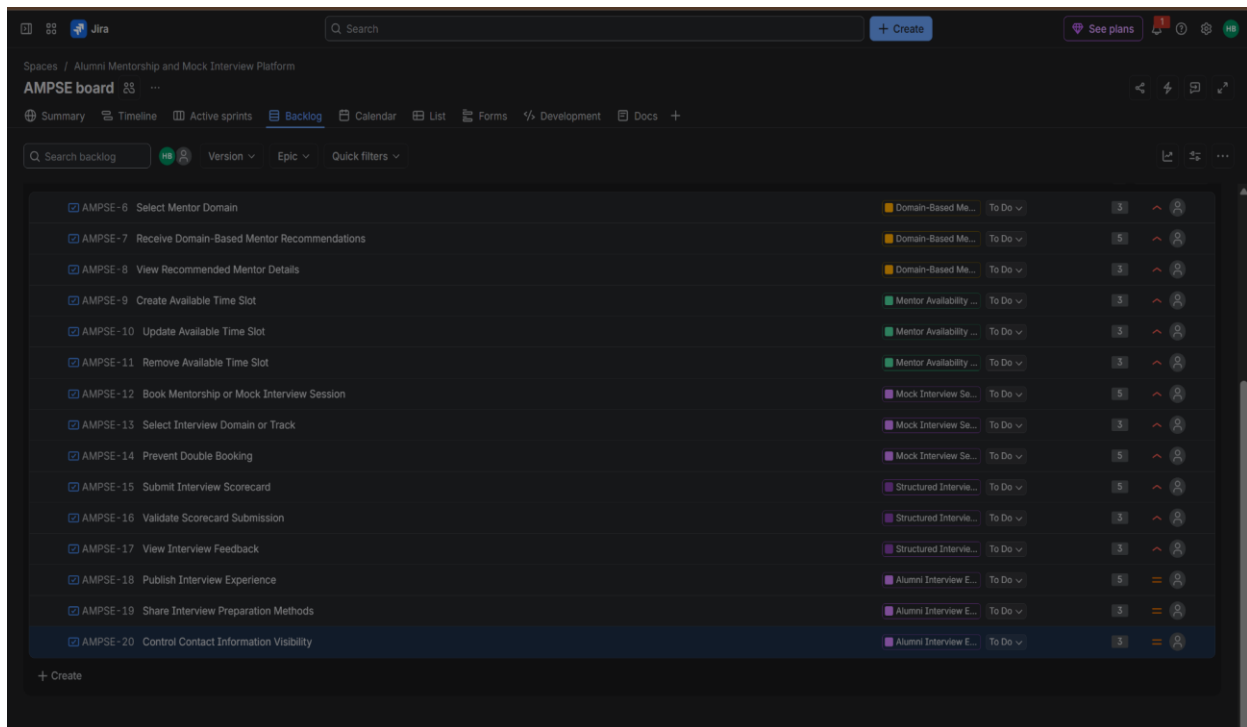


Figure 2 : Jira backlog showing the 15 User Stories, parent Epics, priorities and Story Points.
Total User Stories: 15 ; Total Story Points: 55

4. Requirements Constraints & Traceability

The Non-Functional Requirements were not converted into separate functional Epics. They are carried as cross-cutting constraints in the relevant backlog items.

Lab 1 NFR	Requirement	Lab 2 Traceability
NFR-001	Protect user profile data and feedback scorecards from unauthorized access and support soft-deletion.	Relevant to mentor/profile and scorecard-related stories; privacy and data protection are treated as acceptance constraints.
NFR-002	Alumni contact information shared through interview experience posts shall follow mentor-selected privacy settings.	AMPSE-20 — Control Contact Information Visibility.

5. Sprint Planning & Simulation

Sprint 1 was planned around the main student journey from mentor discovery through session booking and interview-track selection.

Sprint 2 contains the remaining booking protection, assessment, feedback, and interview-experience functionality.

Sprint	Selected Stories	No. of Stories	Story Points	Sprint Goal
Sprint 1	AMPSE-6 to AMPSE-13	8	28	Mentor discovery, availability, session booking and interview-track selection.
Sprint 2	AMPSE-14 to AMPSE-20	7	27	Double-booking prevention, interview assessment, feedback and experience sharing.

5.1 Sprint 1

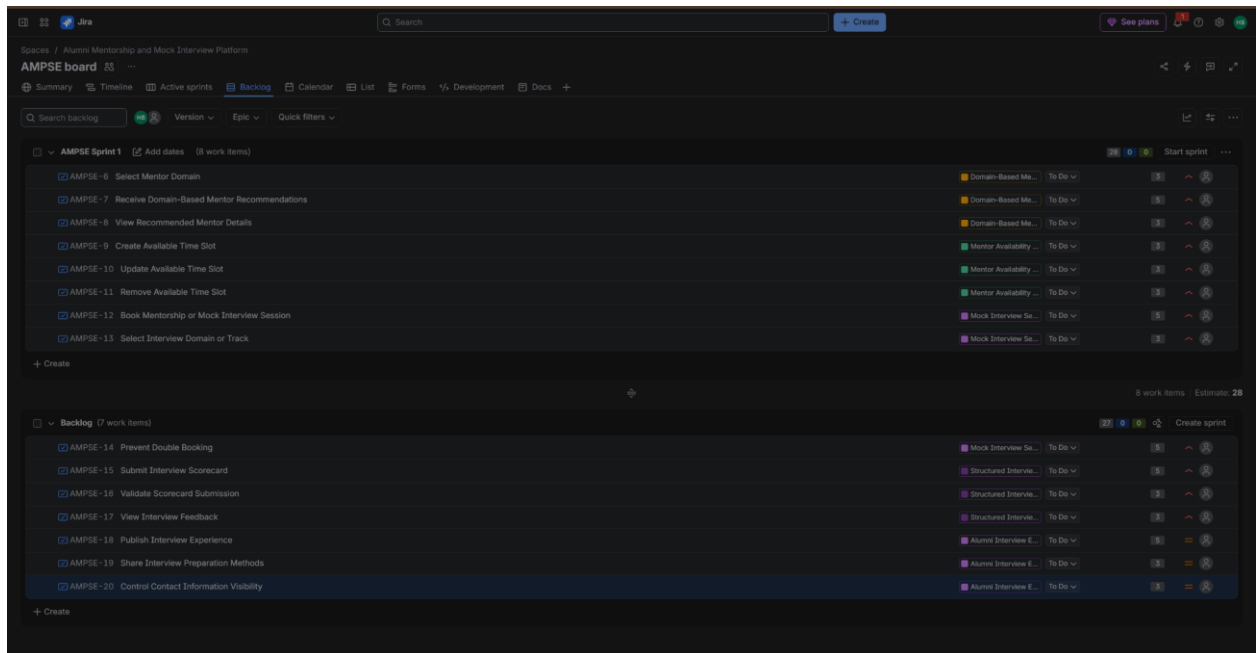


Figure 3 : Sprint 1 planned in Jira with 8 stories and 28 Story Points before starting the sprint.

5.2 Sprint 1 Progress :-

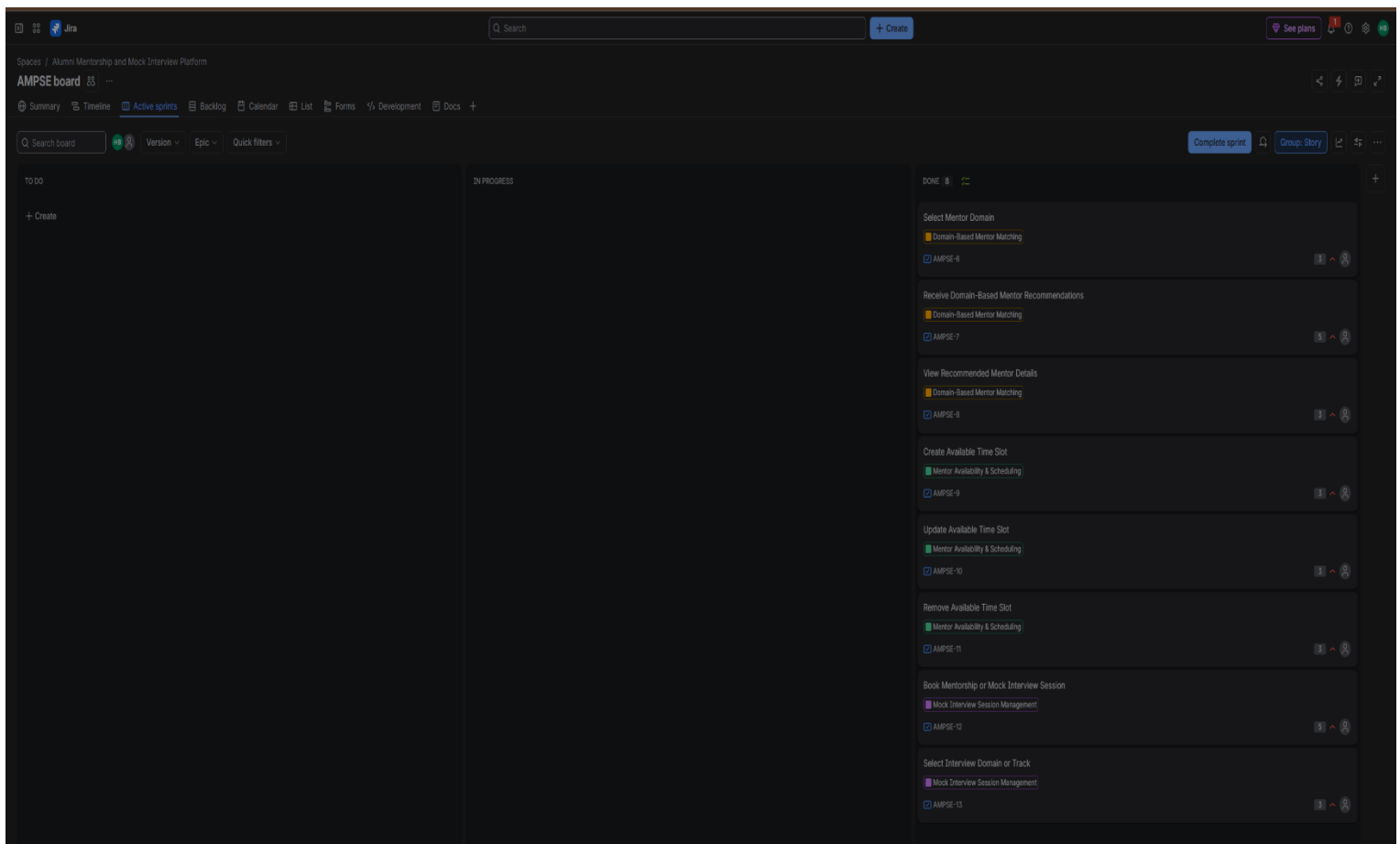


Figure 4 : Sprint 1 completed in Jira with 8 stories and 28 Story Points DONE stage

5.3 Sprint 2

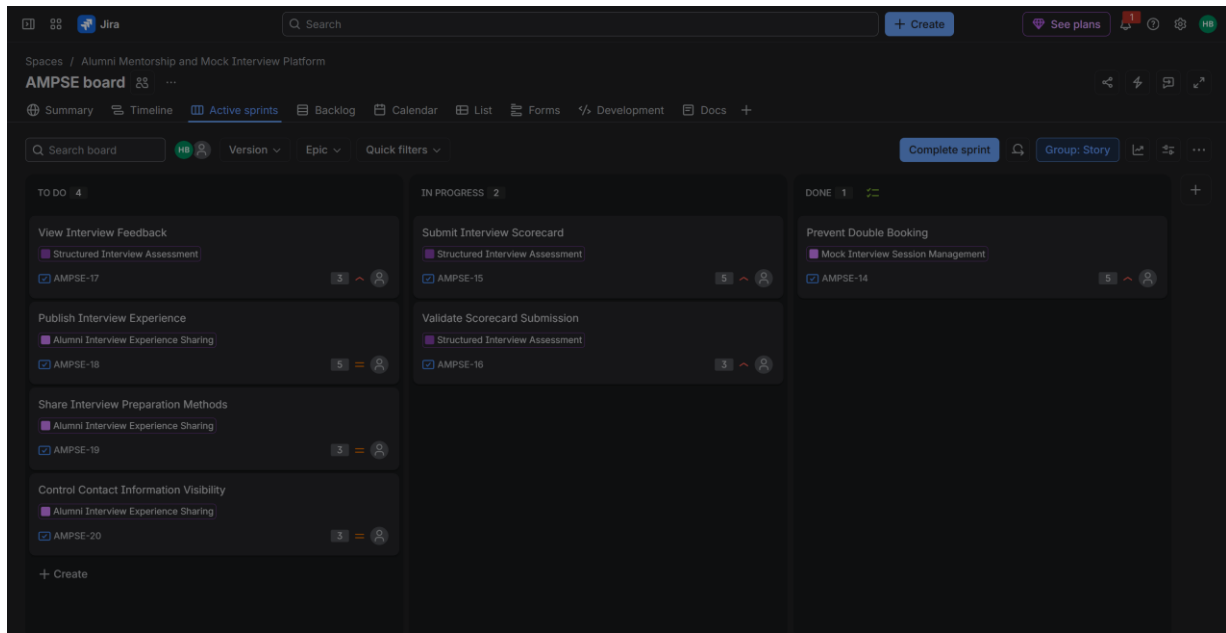


Figure 5 : Sprint 2 planned in Jira with 7 stories and 26 Story Points before starting the sprint.

5.4 Sprint 2 Progress :-

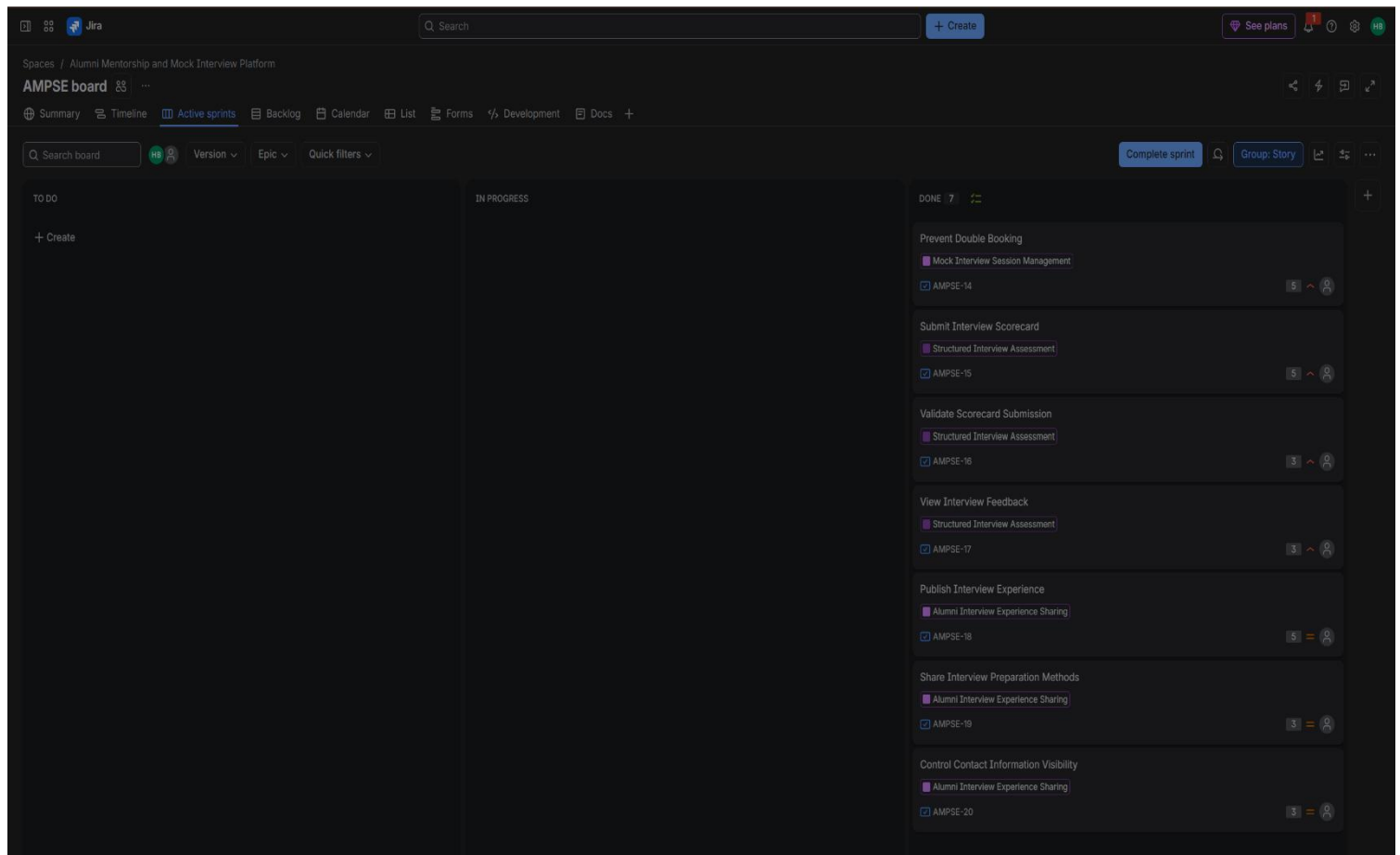


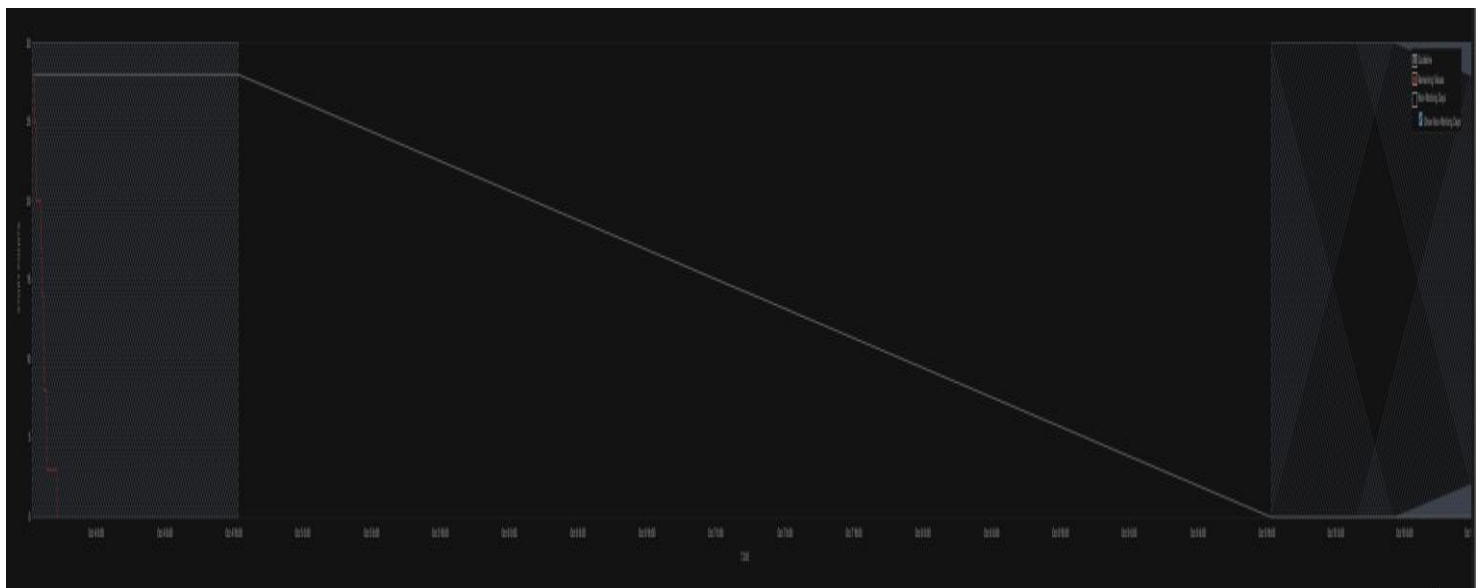
Figure 6 : Sprint 2 completed in Jira with 7 stories and 27 Story Points DONE stage

6. Burndown Chart

The Burndown Chart generated from Jira after the sprint simulation is used to observe the remaining Story Points and the pattern of work completion during the sprint. The chart will be used to observe remaining Story Points across the sprint and to discuss the relationship between planned work and actual simulated progress.



Figure 7 : Burndown Chart for AMPSE Sprint 1 showing progressive reduction in remaining Story Points from 28 to 0.



7. Reflection Questions

1. Did your estimations reflect the actual effort?

The Fibonacci-style Story Points provided a reasonable relative estimate of the effort involved in each User Story. The larger stories, such as receiving mentor recommendations, booking a mentorship or mock interview session, and submitting the interview scorecard, were assigned higher points than smaller tasks such as viewing details or selecting an interview domain. Since the exercise was a Jira simulation, the actual effort cannot be fully validated from elapsed time alone, but the estimates provided a consistent basis for sprint planning.

2. Was your backlog well-prioritized?

Yes. The backlog prioritizes the core functionality of the platform, with mentor discovery, availability, session booking, interview management, and assessment-related functionality receiving higher priority. Supporting functionality such as publishing interview experiences and sharing preparation methods was assigned medium priority, allowing the essential student–mentor workflow to be addressed first.

3. How did your simulated sprint align with your plan?

The sprint plan was followed by dividing the 15 User Stories into two sprints based on functional flow. Sprint 1 focused on mentor discovery, availability, booking, and interview-domain selection, while Sprint 2 covered double-booking prevention, structured assessment, feedback, and interview-experience sharing. Both planned sprints were executed and completed in Jira.

4. What insights did the burndown chart give about your team's capacity?

The Sprint 1 Burndown Chart shows the remaining workload decreasing from **28 Story Points to 0**, with several completion events recorded during the sprint. The chart demonstrates that the planned Sprint 1 workload was completed successfully. However, because this was a simulated sprint and the completion events were concentrated within a short period, the chart should be interpreted as evidence of sprint completion rather than as a precise measurement of real-world team capacity.

8. Conclusion

The Lab 2 exercise transformed the functional requirements identified in Lab 1 for the Alumni Mentorship & Mock Interview Platform into a structured Agile backlog in Jira. Five functional Epics and fifteen User Stories were created, prioritized, and estimated using Fibonacci-style Story Points, resulting in a total backlog of **55 Story Points**. The backlog was divided into two functional sprints, with Sprint 1 containing **8 User Stories and 28 Story Points**, and Sprint 2 containing the remaining **7 User Stories and 27 Story Points**.

Both sprints were planned, executed, and completed using the Jira Scrum board. The Burndown Chart for Sprint 1 showed the reduction of the planned workload from **28 Story Points to 0**, providing a visual representation of work completion during the sprint simulation. Overall, the exercise demonstrated the complete Agile workflow of requirement decomposition, backlog creation, prioritization, estimation, sprint planning, execution, and progress tracking using Jira.